

Fuel Recommendations

⚠ WARNING ⚠

Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.

⚠ CAUTION ⚠

Due to the precise tolerances of diesel injection systems, it is extremely important that the fuel be kept clean and free of dirt or water. Dirt or water in the system can cause severe damage to both the fuel pump and the fuel injectors.

Cummins Inc. recommends the use of American Society of Testing and Materials (ASTM) number 2D fuel. The use of number 2D diesel fuel will result in optimum engine performance.

At operating temperatures below 0°C [32°F], acceptable performance can be obtained by using blends of number 2D and number 1D.

Fuel Recommended Properties

Cummins Inc. recommends the use of fuels that meet the following specifications:

Property	Recommended Specifications
Viscosity (ASTM D445)	1.3 to 4.1 centistokes [1.3 to 4.1 mm per second] at 40°C [104°F].
Cetane Number (ASTM D613)	42 minimum above 0°C [32°F]. 45 minimum below 0°C [32°F].
Sulfur Content (ASTM D129 or D1552)	Not to exceed 5000 ppm on engines without exhaust aftertreatment
Active Sulfur (ASTM D130)	Copper strip corrosion not to exceed number 3 rating after three hours at 50°C [122°F].
Water and Sediment (ASTM D1796)	Not to exceed 0.02 volume-percent on Modular Common Rail System (MCRS) fuel system engines. Not to exceed 0.05 volume-percent on all other High Horsepower engines.

Fuel Recommended Properties	
Carbon Residue (Rams bottom, ASTM D524 or Conradson, ASTM D189)	Not to exceed 0.35 mass percent on 10 volume percent residuum.
Density (ASTM D287)	42 to 30° American Petroleum Institute (API) gravity at 0.816 to 0.876 g/cc at 15°C [60°F].
Cloud Point (ASTM D97)	6°C [10°F] below lowest ambient temperature at which the fuel is expected to operate.
Ash (ASTM D482)	Not to exceed 0.02 mass percent (0.05 mass percent with lubricating oil blending). Oil blending is prohibited on engines with aftertreatment devices.
Distillation (ASTM D86)	10 volume-percent at 282°C [540°F] maximum, 90 volume-percent at 360°C [680°F] maximum, 100 volume-percent at 385°C [725°F] maximum. The distillation curve must be smooth and continuous.
Lubricity High Frequency Reciprocating Rig or Scuffing Load Ball On Cylinder Evaluator	High Frequency Reciprocating Rig: Maximum of 0.52 mm [0.020 in] Wear Scar Diameter at 60°C [140°F]. Scuffing Load Ball On Cylinder Evaluator: Minimum of 3100 grams.
1. In addition to the requirements in the Fuel Recommended Properties table, Cummins Inc. strongly recommends the use of fuel with particle counts less than the International Organization for Standardization (ISO) 4406 code of 18/16/13. See Fluids for Cummins® Products Service Manual, Bulletin 5411406, for more details. 2. For additional information on fuel recommendations and specifications, see Fluids for Cummins® Products Service Manual, Bulletin 5411406. See ordering information in the back of this manual.	

Fuel Cleanliness

This section explains the importance of fuel cleanliness to the successful operation of Cummins® Engines.

Modern fuel systems have been developed to improve combustion performance and fuel consumption, and improve engine performance. These high-pressure fuel systems operate at pressures approaching 2600 bar [37,700 psi], and with component match clearances typically from 2- to 5-microns for injectors. At these pressures, very small, hard particles are potential sources of a fuel system malfunction.

Excessive contamination of diesel fuel can cause premature clogging of diesel fuel filters and/or premature wear of critical fuel injection system parts. Depending on the size and nature of the particles, this can lead to:

- reduced component life
- component malfunction
- fuel system and/or engine malfunction.

Determining fuel cleanliness requires measuring both the size and number of particles per size class in the fuel, i.e. the particle size distribution. The ISO has developed a protocol for expressing the level of contamination by coding the size distribution called ISO 4406. ISO 4406 cleanliness codes are expressed as a series of three numbers (x/x/x), which correspond respectively to the number of particles greater than 4-, 6-, and 14-microns. For example, the numbers in the ISO 4406 rating of 18/16/13 translate to:

- 18 - Up to 2500 particles larger than 4µm (per mL of fuel)
- 16 - Up to 640 particles larger than 6µm (per mL of fuel)
- 13 - Up to 80 particles larger than 14µm (per mL of fuel).

ISO 4406		
ISO Code	Number of Particles per mL	
	More Than	Up to and Including
24	80,000	160,000
23	40,000	80,000
22	20,000	40,000
21	10,000	20,000
20	5000	10,000
19	2500	5000
18	1300	2500
17	640	1300
16	320	640
15	160	320
14	80	160
13	40	80
12	20	40
11	10	20
10	5	10
9	2.5	5
8	1.3	2.5
7	0.64	1.3
6	0.32	0.64
5	0.16	0.32
4	0.08	0.16
3	0.04	0.08
2	0.02	0.04
1	0.01	0.02
0	0.005	0.01
00	0.0025	0.005

Engine builders and fuel injection equipment manufacturers have found that the particles greater than 4-microns and greater than 6-microns are particularly critical to the durability of the fuel injection system. They also recognize that the fuel systems **must** tolerate hard particles smaller than 4-microns that are difficult to filter out, even with the finest filtration. To maximize the efficiency and effectiveness of filtration, Cummins Inc. has adopted the recommendation of the World Wide Fuel Charter that fuel supplied to engines⁽¹⁾ meet the ISO 4406 code of 18/16/13 maximum for respectively 4-, 6-, and 14-micron particle sizes.

Cummins Inc. recommends that if the fuel does **not** meet the ISO cleanliness code of 18/16/13 when supplied to the engine⁽¹⁾, additional filtration be applied before the fuel is delivered to the equipment's fuel tank. A Cummins® Distributor or Cummins Filtration™ representative can supply hardware and additional filtration guidance and can recommend countermeasures such as improved fuel quality from the fuel supplier, and/or better fuel handling, storage, dispensing, and fuel tank cleaning techniques.

1. Fuel quality **must** be assessed at the inlet to the suction side fuel filter.

For the requirements for Using Biodiesel Fuel in Cummins® Engines, use the following for details on acceptable biodiesel blends. See Fluids for Cummins® Products Service Manual, Bulletin 5411406.